Deep-Learning-Enabled Data Synchronization for Noninteger Time Lags in Wireless Structural Health Monitoring

Xiao Yu, Yuguang Fu, Xuewen Yu, and Shuaiwen Cui

***Abstract*—Accurate synchronization of distributed sensor measurements is essential for reliable vibration-based structural health monitoring (SHM). Wireless Internet of Things (IoT) systems, while attractive for large-scale SHM deployments, inevitably introduce unknown time lags arising from clock drift and communication delays. Conventional frequency-domain (FD) estimators and feature-based machine learning (ML) models struggle to handle such asynchronous measurements, especially when delays are smaller than the sampling interval. To address these limitations, this article proposed a learning-based synchronization framework that integrates a time-lag convolutional neural network with attention (TLCA) to directly estimate continuous-valued time lags from raw vibration signals, together with a resampling-based alignment strategy to correct noninteger delays. The proposed framework was validated using both simulated structural responses and field data from a wireless SHM system deployed on the Jindo Bridge. By directly learning time relationships from raw vibration signal pairs, TLCA achieves submillisecond accuracy and consistently outperforms conventional FD and feature-based learning methods, with about an order-of-magnitude improvement in precision. It also maintained strong robustness under increased noise and modal coupling. With efficient data-level alignment and low computational complexity of the overall framework, it can enable accurate and real-time synchronization for practical IoT-based SHM applications.**

***Index Terms*—Data synchronization, end-to-end deep learning (DL), noninteger time lag, structural health monitoring, wireless Internet of Things (IoT).**

I. Introduction

STRUCTURAL health monitoring (SHM) aims to ensure the safety and long-term performance of civil infrastructure by continuously evaluating structural integrity through sensing and data analysis. To support large-scale and long-term monitoring, modern SHM systems increasingly adopt distributed sensing architectures based on wireless Internet of Things (IoT) technologies [1]. Such architectures enable spatially dense measurements while offering improved scalability, reduced cost, and efficient deployment [2]. Within these systems, vibration signals collected by edge devices serve as a primary data source for structural assessment [3]. By jointly analyzing vibration responses collected from multiple spatial locations, key modal properties, including natural frequencies, damping ratios, and mode shapes, can be identified [4], which provide vital insight into the dynamic behavior and potential degradation of structures [5], [6]. However, in the distributed sensing architecture, each sensor node operates with an independent local clock for timekeeping, which is subject to hardware imperfections and environmental variations [7]. This inevitably leads to different temporal drift and accumulates over time, resulting in asynchronous signals across the network [8]. Moreover, vibration signals are acquired as discrete-time samples, making noninteger time lags unavoidable when misalignment occurs between sensor measurements. Such temporal misalignment introduces phase errors that can severely distort modal analysis and undermine the reliability of damage detection [9]. This issue is particularly critical in wireless SHM, where structural acceleration responses are recorded at high sampling rates overextended durations [10]. Even submillisecond time misalignment can introduce significant phase errors; for example, a 1-ms offset corresponds to a 3.6° phase error at 10 Hz and a 36° phase error at 100 Hz, which can result in erroneous damage assessment [11]. As a result, SHM imposes stringent requirements on synchronization accuracy across distributed sensor measurements [12].

Clock synchronization protocols align local clocks across distributed sensor nodes through beacon exchanges or reference broadcasts, thereby providing network-wide timestamp consistency. Representative protocols include the timing-sync protocol for sensor networks (TPSNs) [13], reference broadcast synchronization (RBS) [14], and flooding time synchronization protocol (FTSP) [15]. However, these protocols still cannot guarantee synchronized sensing because variations in communication latency, analog-to-digital converter (ADC) triggering jitter, sensor readout timing, and operating-system scheduling introduce residual sampling offsets in the recorded data [16], [17], [18]. As a result, system-level clock synchronization alone is insufficient for high-precision vibration analysis, necessitating post hoc signal-level alignment to correct the remaining time delays in the recorded data.

Post hoc signal-level synchronization consists of time-lag estimation followed by signal alignment. Traditional

Received 25 December 2025; revised 25 January 2026; accepted 12 February 2026. Date of publication 16 February 2026; date of current version 11 May 2026. This work was supported in part by the Ministry of Education (MOE) Academic Research Fund (AcRF) Tier 1 under Grant RG146/23, in part by the College of Engineering (CoE) Dean's Interdisciplinary Grant in Nanyang Technological University Grant 03INS002320C120, and in part by the Nanyang Technological University (NTU) Start-Up Grant 03INS001210C120. *(Corresponding author: Yuguang Fu.)*

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Digital Object Identifier 10.1109/JIOT.2026.3664904

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approaches include time-domain methods such as the autoregressive moving average vector (ARMAV) model [19] and stochastic subspace identification (SSI) [20], as well as frequency-domain (FD) techniques that estimate time lags from the slopes of cross-spectral density (CSD) phase spectra [21], [22], [23]. However, these methods depend on prior knowledge (e.g., model order selection or bandwidth specification), making their performance sensitive to user experience and adversely affecting performance. They also rely on simplifying assumptions, such as zero fundamental-mode phase and linear structural damping, which are rarely satisfied in real-world SHM deployments. Furthermore, these methods treat time lags as integer delays and align signals through direct index shifting, limiting the accuracy to the time interval and neglecting noninteger (subsample) delays that cannot be corrected by sample shifting alone. Resampling-based alignment methods [12], [16], [24] consider noninteger time lags, but they rely on beacon exchanges and node-reported timestamps to align signals, making their accuracy constrained by timestamp precision. The frequent communication increases energy consumption and disrupts ongoing sensing tasks. To date, no existing method performs synchronization directly from vibration signals while handling noninteger time lags.

Learning-based approaches have emerged as a powerful tool in signal processing and SHM, demonstrating strong capability in feature extraction and modeling complex temporal dependencies. They have been successfully applied to damage identification [25], [26], [27], modal analysis [28], [29], [30], and data anomaly detection [31], [32], [33], offering improved robustness under noise, and environmental variability. However, no existing work applied deep learning (DL) to time-lag estimation in wireless SHM systems. From a physical SHM perspective, time-lag estimation corresponds to learning a complex phase–time mapping induced by modal superposition and signal shift, which motivates the use of DL as a data-driven estimator rather than a replacement of the underlying vibration model.

Residual data-level time lags persist even after clock synchronization and inherently include nonintegers in nature. Existing signal processing approaches require prior experience and are often limited to integer-delay accuracy. Resampling methods can correct noninteger time lags, but they depend on timestamp information. These limitations motivate the development of a data-driven, timestamp-free synchronization framework that can estimate continuous-valued time lags and enable data-level alignment. To bridge this gap, this article proposed a complementary DL-based synchronization framework for wireless SHM systems. In the first stage, a time-lag convolutional neural network with attention (TLCA) is introduced to estimate continuous time lags directly from raw vibration signals. By leveraging the representational capacity of DL, TLCA captures hierarchical temporal dependencies across sensor signals, reducing reliance on manual feature engineering, prior expertise, and overly simplified analytical assumptions. The second stage proposed a Bayesian-optimized (BO) resampling strategy to achieve noninteger synchronization based on the TLCA predicted time lags, enabling accurate subsample synchronization without reliance on timestamps. The complete framework was validated using simulated building responses and field data from the Jindo Bridge, demonstrating superior accuracy across both small and large time lags compared to state-of-the-art (SOTA) synchronization methods.

The main contributions of this study are given as follows.

1) To address the limitation of conventional time-lag estimation methods in handling noninteger time lags, a data-driven TLCA model was developed to estimate continuous-valued time lags directly from raw vibration signals, leading to improved submillisecond accuracy and robustness over SOTA approaches.
2) To overcome the reliance of existing noninteger alignment methods on node timestamps and beacon exchanges, a BO-assisted resampling strategy was introduced to enable precise subsampling and data-level synchronization and reduce communication overhead.
3) To support practical deployment on resource-constrained edge devices, the proposed framework was designed to be compact and computationally efficient, with 108k trainable parameters and submillisecond inference latency, making it well suited for IoT-based SHM applications.

The remainder of this article is organized as follows. Section II formulates the physical and mathematical model of the synchronization problem in wireless SHM, with emphasis on noninteger delays. Section III introduces the proposed DL-based synchronization framework, including the TLCA architecture and the BO resampling strategy. Section IV describes the simulated and field datasets along with the data synthesis and preprocessing procedures. Section V presents the experimental setup and baseline models. Section VI analyzes results and provides discussions. Finally, Section VII concludes this article and outlines future directions.

II. System Model and Problem Formulation

This section establishes the physical and mathematical foundations of the synchronization problem and noninteger time lags.

A. Measurement Model and Noninteger Time-Lag Definition

The dynamic response of a civil structure subjected to ambient or operational excitation can be approximated by an N_d-degree-of-freedom (DOF) model expressed as

$$\mathbf{M}\ddot{\boldsymbol{\xi}}(t) + \mathbf{C}\dot{\boldsymbol{\xi}}(t) + \mathbf{K}\boldsymbol{\xi}(t) = \mathbf{P}(t) \tag{1}$$

where $\mathbf{M}, \mathbf{C}, \mathbf{K} \in \mathbb{R}^{N_d \times N_d}$ denote the mass, damping, and stiffness matrices, respectively. The structural displacement state vector is $\boldsymbol{\xi}(t) = [\xi_1(t), \xi_2(t), \ldots, \xi_{N_d}(t)]^T$, with $\dot{\boldsymbol{\xi}}(t)$ and $\ddot{\boldsymbol{\xi}}(t)$ representing the corresponding velocity and acceleration vectors. $\mathbf{P}(t) \in \mathbb{R}^{N_d}$ denotes the external excitation. N_d denotes the number of structural DOF.

In a wireless sensor network deployed on the structure, vibration signals collected from spatially distributed sensor nodes form the continuous-time observation vector

$$\mathbf{y}(t) = [y_1(t), y_2(t), \ldots, y_{N_s}(t)]^T \tag{2}$$

where N_s denotes the number of sensors.

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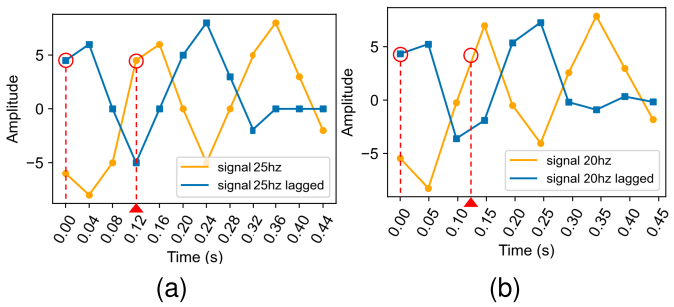


Fig. 1. Illustration of noninteger time lag. (a) Signals sampled at 25 Hz with 0.12-s time lag. (b) Signals sampled at 20 Hz with 0.12-s time lag.

For sensor i, the vibration signal is represented by the continuous-time signal $y_i(t)$, defined over the observation interval $t \in [0, T_{\text{obs}}]$.

Sampling at a rate $f_s = 1/T_s$, where T_s is the sampling uniform interval, the discrete-time measurement is

$$y_i[n] = y_i(t)\big|_{t=nT_s}, \quad n = 0, 1, \ldots, L-1 \tag{3}$$

where L denotes the total number of samples.

Accordingly, stacking the sampled measurements from all sensor nodes yields the discrete-time observation vector

$$\mathbf{y}[n] = [y_1[n], y_2[n], \ldots, y_{N_s}[n]]^T. \tag{4}$$

Consider two sensor channels $y_i(t)$ and $y_j(t)$ extracted from $\mathbf{y}(t)$ with a time lag τ, and the aligned signal is defined as

$$y'_j(t) = y_j(t - \tau). \tag{5}$$

where $y'_j(t)$ denotes the time-shifted version of $y_j(t)$ aligned to the time axis of $y_i(t)$.

τ is a continuous time lag and can be decomposed as

$$\tau = kT_s + \delta, \qquad 0 \le \delta < T_s \tag{6}$$

where $k \in \mathbb{Z}$ is the integer number of samples (i.e., the number of full sampling intervals) and δ represents the fractional (subsample) time lag.

When $\delta = 0$, the required time instant coincides with the sampling grid

$$nT_s - \tau = (n-k)T_s \quad \Rightarrow \quad y'_j[n] = y_j[n-k]. \tag{7}$$

When $\delta > 0$, the required time instant no longer lies on the sampling grid

$$(n-k)T_s - \delta \notin \{mT_s \mid m \in \mathbb{Z}\} \tag{8}$$

indicating a fractional (noninteger) delay that requires subsample alignment and direct time shifting leads to waveform misalignment. As illustrated in Fig. 1, a time lag of 0.12 s corresponds to three samples at a sampling rate of 25 Hz, whereas the same lag falls between sampling instants at 20 Hz and thus becomes noninteger. The acquired sensor data in SHM inherently represent the real-world dynamic behavior of structures and carry physical significance. Although sampled discretely in the time domain, their FD characteristics remain consistent across different sampling rates provided the Nyquist criterion is satisfied [34]. Consequently, simple index shifting is insufficient for accurate synchronization, and arbitrary sample insertion or deletion is inappropriate.

B. Problem Formulation

The synchronization problem consists of time-lag estimation followed by signal alignment.

1) Time-Lag Estimation: Consider two discrete-time vibration signals $y_i[n]$ and $y_j[n]$ acquired by spatially distributed wireless sensor nodes with the underlying continuous-time lag τ.

The objective of time-lag estimation is to infer the unknown time lag $\tau \in \mathbb{R}$ from the paired asynchronous measurements, which is formulated as a supervised regression problem

$$\hat{\tau} = f_\theta\left(y_i[n], y_j[n]\right), \quad n = 0, 1, \ldots, L-1 \tag{9}$$

where $f_\theta(\cdot)$ denotes a DL model with learnable parameters θ and $\hat{\tau}$ is the estimated time lag.

The model parameters are trained by minimizing the estimation error between the predicted time lag $\hat{\tau}$ and the ground truth τ. Specifically, the training objective is defined as

$$\theta^* = \arg\min_\theta \ \mathbb{E}\left[\mathcal{L}\left(f_\theta\left(y_i[n], y_j[n]\right), \tau\right)\right] \quad n = 0, 1, \ldots, L-1 \tag{10}$$

where $\mathcal{L}(\cdot)$ denotes a regression loss function, and the expectation is taken over the training data distribution.

During inference, the trained model estimates the unknown time lag for previously unseen signal pairs according to

$$f_\theta : \left(\{y_i[n]\}_{n=0}^{L-1}, \{y_j[n]\}_{n=0}^{L-1}\right) \to \hat{\tau}. \tag{11}$$

Within this formulation, the proposed DL model is trained to learn a mapping between paired asynchronous vibration signals and their corresponding time lag τ and acts as a data-driven time-lag estimator that implicitly captures the phase–time relationships under measurement noise.

2) Signal Alignment: Given the estimated continuous-valued time lag $\hat{\tau}$, the alignment stage aims to construct a time-aligned version $y'_j[n]$ of the asynchronous signal $y_j[n]$ while preserving its physical and spectral characteristics. This can be formulated as

$$y'_j[n] = \arg\min_{z[n]} \ \mathcal{L}\left(y_i[n], z[n]\right) \quad \text{s.t. } z[n] = \mathcal{A}\left(y_j[n], \hat{\tau}\right) \quad n = 0, 1, \ldots, L-1 \tag{12}$$

where $\mathcal{A}(\cdot)$ denotes a fractional-delay alignment operator (implemented via resampling) and $\mathcal{L}(\cdot)$ measures mismatch between the aligned signal and the reference signal.

3) Physical Assumptions: The formulation is established under the following assumptions.

1) *Uniform Sampling Interval:* Each sensor node samples vibration signals with a constant sampling interval T_s over the observation window.
2) *Homogeneous Sampling Rates:* All sensor nodes operate at the same nominal sampling rate f_s.
3) *Linear Structural Dynamics:* The structure is linear and time-invariant over the observation window.

These assumptions enable a clear formulation of the synchronization problem and allow the DL model to focus on the learning phase–time relationships induced by signal misalignment rather than confounding sampling irregularities.

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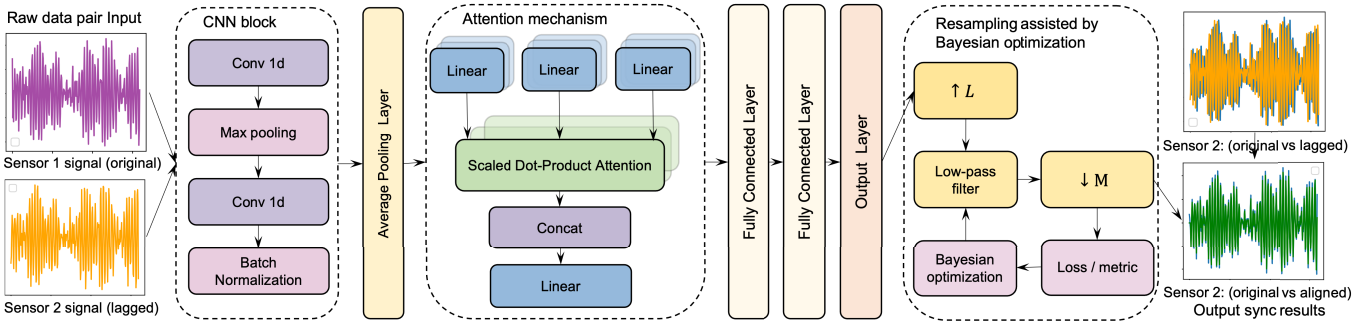


Fig. 2. Architecture of the time-lag estimation model TLCA.

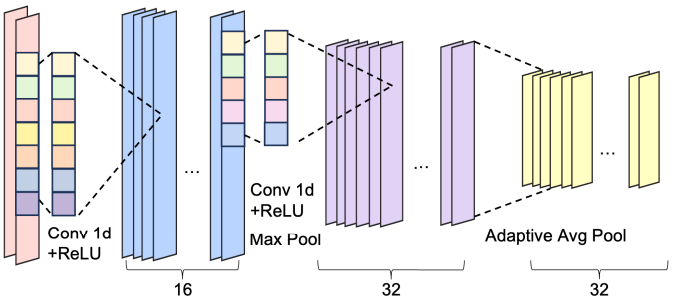


Fig. 3. Architecture of the CNN block.

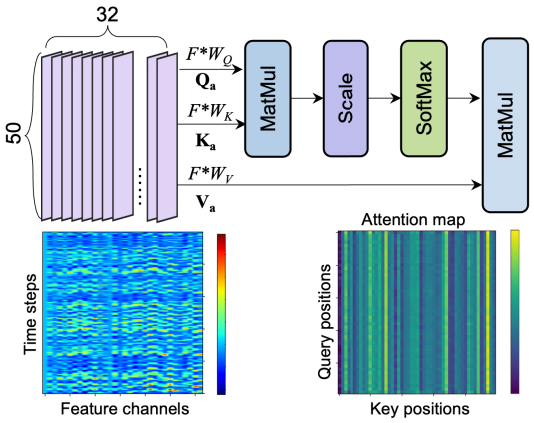


Fig. 4. Architecture of the attention block.

III. Methodology

Even a small deviation can significantly affect mode shapes that are related to vibration and damage analyses, especially in large-scale and complex structural systems. Although numerous algorithms have been proposed to synchronize sensing, their performance is limited to integer time lags. This study proposed a synchronization framework as shown in Fig. 2 that integrates a DL model, TLCA, for time-lag estimation with a resampling method assisted by BO, enabling precise signal alignment.

A. Time-Lag Estimation Model

The proposed time-lag estimation model TLCA integrates a convolutional neural network (CNN) block, an attention mechanism, and fully connected layers. The CNN is effective in capturing local features such as waveform variations and short-term dependencies. The attention learns long-range dependencies and spatial relationships across sensor signals, enabling adaptive highlighting of critical feature channels and significantly enhancing representational capacity. Finally, fully connected layers integrate extracted features and regress a continuous time-lag value. The inputs are pairs of 1-D time-series signals recorded by two sensors on the same structure with an unknown time lag. It is an end-to-end, data-driven DL model trained directly on raw data, avoiding reliance on handcrafted features and assumptions.

1) CNN Block: The raw signal pairs are first processed by a lightweight CNN block, which serves as a foundational feature extraction module in the proposed model architecture, as shown in Fig. 3. The two signals are stacked as a two-channel input and jointly processed by the CNN. The block consists of two sequential 1-D convolutional layers with 16 and 32 output channels. The first layer employs a kernel of 7 to capture local oscillatory patterns, followed by max pooling to reduce temporal resolution. The second convolution further refines higher level temporal features, with batch normalization. Finally, average pooling reduces feature dimensionality before passing to the attention block, and the raw vibration signals are transformed into high-dimensional feature maps that encode delay-sensitive temporal structures.

2) Attention Block: The features extracted by CNN are then processed by a self-attention block, which adaptively reweights temporal features according to their relevance to synchronization. As illustrated in Fig. 4, the features are linearly projected by the learned weight matrices $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$ to generate the query ($\mathbf{Q_a}$), key ($\mathbf{K_a}$), and value ($\mathbf{V_a}$) representations. The projection weights are learnable parameters and act as feature selectors optimized during training via backpropagation. In the context of vibration-based SHM, each temporal position in the $\mathbf{K_a}$ represents a candidate reference feature while $\mathbf{Q_a}$ seeks synchronization-relevant information. The similarity between the $\mathbf{Q_a}$ and $\mathbf{K_a}$ in (13) reflects phase-consistent and delay-related correlations between vibration features at different time instants. The resulting attention weights highlight temporal regions that contribute most to time-lag estimation. After scaling by $\sqrt{d_f}$ and normalization via a softmax function, the weighted aggregation of the value matrix $\mathbf{V_a}$ yields an enhanced feature representation

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that emphasizes synchronization-relevant vibration components while suppressing noise and less informative patterns. The attention-enhanced features are subsequently passed to fully connected layers for regression to estimate continuous-valued time lags

$$\text{Attention}(\mathbf{Q_a}, \mathbf{K_a}, \mathbf{V_a}) = \text{softmax}\left(\frac{\mathbf{Q_a}\mathbf{K_a}^\top}{\sqrt{d_f}}\right)\mathbf{V_a} \tag{13}$$

where d_f is the feature dimension of the query and key vectors.

During training, paired vibration signals with known time lags are used as supervised samples. Each signal pair is normalized and segmented into fixed-length windows before being fed into the TLCA model. The network is trained to minimize the Huber loss between the predicted and ground-truth time lags using the Adam optimizer. Training is performed for a fixed number of epochs with early stopping based on validation loss. During inference, a pair of asynchronous vibration signals is directly fed into the trained TLCA model to estimate the continuous-valued time lag, and it is subsequently used by the BO-assisted resampling module to perform noninteger signal alignment at the data level.

B. Synchronization and Bayesian Optimization

Accurate synchronization under noninteger time lags requires estimating intermediate values between discrete samples. To this end, this study adopts a resampling strategy assisted by BO to achieve finer synchronization while preserving frequency content and phase relationships. The signal $y_j[n]$ sampled at f_s is first upsampled by inserting zeros between adjacent samples, thereby expanding the effective time grid, defined in (14). A low-pass anti-aliasing filter is then applied to suppress spectral components beyond the new Nyquist frequency and to remove artifacts introduced during upsampling, as formulated in (15). The continuous waveform is subsequently reconstructed via convolution with an interpolation kernel and shifted according to the estimated time lag $\hat{\tau}$ with subsample precision. Finally, the signal is downsampled back to its original resolution, yielding a temporally aligned signal $y'_j[n]$ while maintaining the essential FD characteristics. Unlike traditional resampling methods that rely on timestamps to address start-up time uncertainty [34], the proposed approach synchronizes signals directly from data using time lags estimated from the DL model, eliminating dependence on timestamp information.

In the data resampling synchronization, selecting an appropriate upsampling frequency is critical. While higher rates provide finer temporal resolution, they also increase computational cost and may introduce numerical distortion. To balance accuracy and efficiency, BO is employed to automatically determine the optimal upsampling frequency f_o.

The mathematical formulation of the resampling and optimization procedures is summarized as follows.

1) Resampling Formulation: The upsampled signal $y_{j,\uparrow}[n]$ is defined as

$$y_{j,\uparrow}[n] = \begin{cases} y_j\left[\dfrac{n}{U}\right], & n = \ell U,\ \ell \in \mathbb{Z} \\ 0, & \text{otherwise} \end{cases} \tag{14}$$

where $U = f_s^{\uparrow}/f_s \in \mathbb{N}$ denotes the upsampling factor.

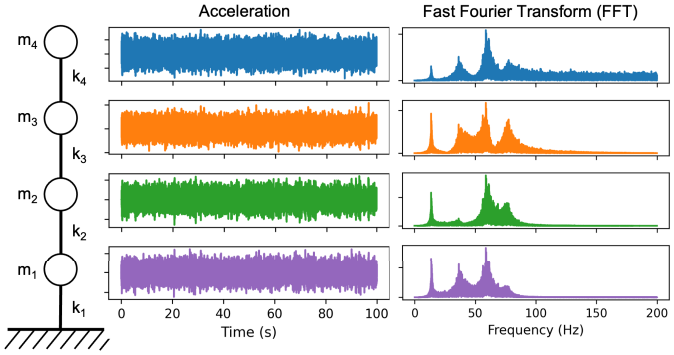


Fig. 5. Time and frequency domain of representation of the 4-DOF structure.

The filtered signal is obtained by

$$\tilde{y}_j[n] = (y_{j,\uparrow} * h_r)[n] \tag{15}$$

where $*$ denotes the convolution and $h_r[n]$ is a low-pass anti-aliasing filter for resampling.

2) Bayesian Optimization Formulation: To quantify the synchronization quality under a given upsampling frequency $f_s^{\uparrow}$ (Hz), an objective function is defined as

$$J\left(f_s^{\uparrow}\right) = \alpha\,\rho\left(f_s^{\uparrow}\right) + \beta\,\gamma\left(f_s^{\uparrow}\right) \tag{16}$$

where $\rho(\cdot)$ is the Pearson correlation coefficient, $\gamma(\cdot)$ is the maximum cross correlation, and α and β are weighting coefficients.

Following a standard BO framework [35], a probabilistic surrogate model is iteratively updated using previously evaluated pairs $\left(f_s^{\uparrow}, J(f_s^{\uparrow})\right)$. An acquisition function $A(\cdot)$ is then employed to select the next candidate upsampling frequency according to

$$f_o = \arg\max_{f_s^{\uparrow} \in \mathcal{F}} A\left(f_s^{\uparrow}\right) \tag{17}$$

where $\mathcal{F}$ denotes the predefined search space of candidate upsampling frequencies and $A(\cdot)$ balances exploration and exploitation. The optimization procedure continues until convergence or a predefined stopping criterion is met.

IV. Datasets Description

To evaluate the proposed TLCA model and synchronization method, two datasets were considered: 1) simulated responses generated from a four DOF benchmark structure and 2) field vibration data from the Jindo Bridge. The details of each dataset and data synthesis procedures are presented as follows.

A. Description of the Simulated Dataset

The simulated dataset was generated using a 4-DOF shear-building model, where each floor is represented by a lumped mass with a single lateral translational DOF, as shown in Fig. 5. The dynamic response is governed by the equation of motion in (1). The masses and stiffness values were selected to match the approximate dynamic characteristics of the laboratory structure: $M = [7.5, 7.5, 6.1, 5.1]$ kg and $K = [340, 450, 420, 330]$ kN/m. Nonproportional modal damping was applied by assigning mode-specific damping

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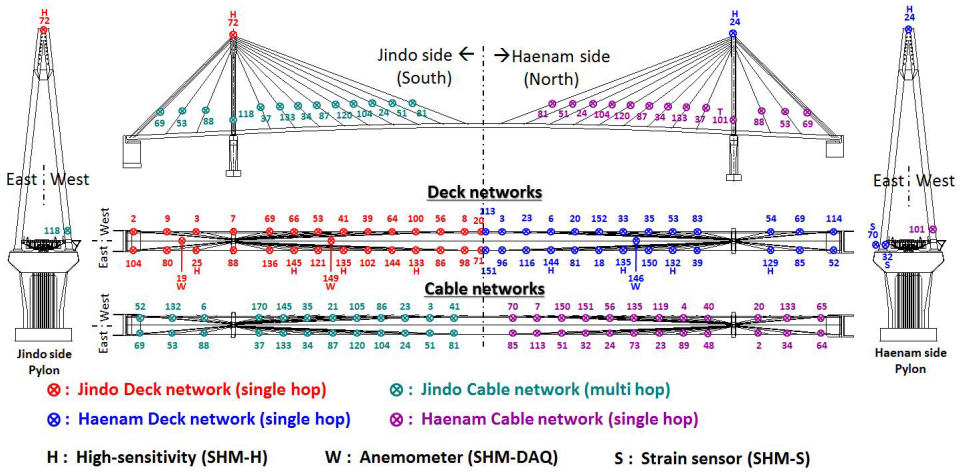


Fig. 6. Spatial distribution of sensor nodes deployed on the Jindo Bridge [39].

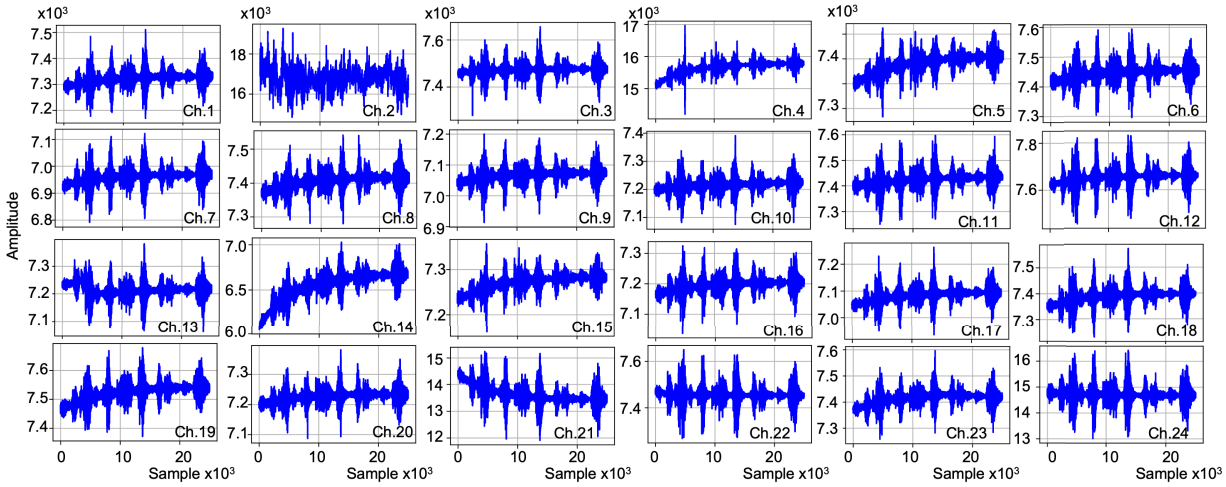


Fig. 7. Data recorded from 24 sensors on the Jindo Bridge.

ratios to the first four vibration modes. These damping ratios were converted to a physical damping matrix through modal transformation, producing $C = \Phi^{-T} C_m \Phi^{-1}$, consistent with standard structural dynamics procedures. The structural parameters were selected based on real-world building properties to ensure physical validity. A band-limited random excitation (20–150 Hz) was used to excite the structural modes while avoiding low-frequency rigid-body responses. The acceleration responses were simulated using the state space model. Four virtual accelerometers sampled the acceleration responses at 400 Hz for 100 s, producing 40 000 samples per channel. Representative time histories and their fast Fourier transform (FFT) spectra are shown in Fig. 5. The resulting signals were fully synchronized and provide a controlled yet physically representative dataset for evaluating vibration-based SHM algorithms.

B. Field Dataset From the Jindo Bridge

The second dataset was collected from the Jindo Bridge, a twin cable-stayed bridge in South Korea [36], [37]. The second Jindo Bridge consists of a 344-m main span and two 70-m side spans, supported by parallel-strand stay cables anchored to diamond-shaped pylons. A large-scale wireless smart sensor network, comprising 70 Imote2-based sensor nodes, was deployed across the deck, pylons, and cables (Fig. 6). The wireless sensing platform was based on the Imote2, with each equipped with tri-axis accelerometers, strain gauges, and environmental sensors [38]. Clock synchronization was handled by the FTSP, which disseminates timestamps through the network in a flooding manner for millisecond-level alignment. Each receiving node adjusts its local clock based on the embedded timestamp and compensates for clock drift through linear regression. Field implementations on the bridge have demonstrated that, immediately after installation and calibration, Imote2 nodes provided synchronized data streams across the network [37]. This provides a reliable baseline for validating data-driven synchronization methods against ground-truth synchronized field measurements. Each dataset consists of 1000 s of acceleration responses at a sampling frequency of 25 Hz. Fig. 7 shows acceleration responses collected from 24 sensors over the same duration. The field dataset captures the complexities of large-scale

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infrastructure with substantial environmental noise in contrast to the relatively clean and controlled signals of simulated structures.

C. Data Preprocessing

Given that $y_i[n]$ and $y_j[n]$ are the discrete-time acceleration response measured by sensor i and j, where n is the discrete-time index and the signal is sampled at a rate f_s (Hz), for each sensor pair (i, j), a valid starting index and a discrete time lag k_r were randomly drawn from predefined ranges. The equal length of L samples signal segments was then extracted to ensure waveform continuity with a known relative time shift. The original unshifted segments were retained as ground truth, whereas the time-shifted segments simulated realistic integer delays. After extraction, all segments were low-pass filtered using a finite impulse response (FIR) anti-aliasing filter and subsequently downsampled to a lower sampling rate f_s' (Hz). Since the time offsets were applied at the original high sampling rate, the resulting delays became fractional relative to the lower sampling grid, thereby introducing controlled noninteger time lags (Algorithm 1). This preprocessing strategy preserves the physical continuity and spectral characteristics of vibration responses while enabling the generation of both integer and noninteger time lags representative of practical desynchronization scenarios in wireless SHM networks.

For the simulated dataset, each extracted segment contained $L = 4000$ samples recorded at $f_s = 400$ Hz and was downsampled to $f_s' = 100$ Hz. The maximum discrete time lag was set to $k_{\max} = 200$ samples at 400 Hz, corresponding to a maximum continuous time lag of $\tau_{\max} = 0.5$ s. Time lags were sampled with a resolution of 0.0025 s, yielding subsample misalignments finer than the 0.01-s resolution associated with the 100-Hz sampling rate. For the field dataset, each segment consisted of $L = 2000$ samples originally recorded at 25 Hz and downsampled to 20 Hz. A maximum lag of 0.4 s was imposed, and additional offsets with increments of 0.04 s were introduced to generate controlled noninteger delays. The resulting asynchronous signal pairs were used as supervised inputs to train and evaluate the proposed TLCA model [Fig. 7(b)]. The complete dataset was partitioned into training, validation, and testing subsets using a 7:1:2 ratio.

Algorithm 1 Synthesis of Noninteger Time-Lagged Dataset

Input: Vibration dataset $\mathcal{D} = \{y_1[n], \ldots, y_{N_s}[n]\}$, collected by N_s spatially distributed sensors, sampled at rate f_s (Hz)

Output: Dataset $\mathcal{D}'$ containing asynchronous signal pairs $(\tilde{y}_i[n], \tilde{y}_j^{(\tau)}[n])$ and the corresponding time-lag set $\{\tau\}$

Initialize: $\mathcal{D}' \leftarrow \emptyset$; segment length L (number of discrete samples); target down-sampling rate f_s' (Hz); discrete time-lag distribution $\mathcal{P}_k(k_{\min}, k_{\max})$ (number of samples)

Design an FIR anti-aliasing low-pass filter $h_{\text{lp}}[n]$ with cutoff frequency $f_c \le \frac{1}{2} f_s'$ (Hz) and 101 taps

for each synthesis iteration $r = 1$ to R **do**

Randomly select two distinct sensors $i, j \sim \{1, \ldots, N_s\}$, $i \neq j$

Randomly select a discrete time lag $k_r \sim \mathcal{P}_k(k_{\min}, k_{\max})$ (samples) and **the continuous time lag corresponds to** $\tau_r = k/f_s$ (s)

Select a valid starting index $n_i \sim \{0, \ldots, T - L\}$ where T denotes the total number of samples available in each original vibration signal $y_i[n]$.

Compute the shifted starting index

$$n_j = n_i + k$$

if $n_j < 0$ **or** $n_j + L > T$ **then**

continue ▷discard invalid segment

end if

Extract signal segments

$$\hat{y}_i = y_i[n_i : n_i + L], \quad \hat{y}_j = y_j[n_j : n_j + L]$$

Low-pass filter and down-sample

$$\tilde{y}_i[n] = (\hat{y}_i * h_{\text{lp}})[n] \downarrow f_s', \quad \tilde{y}_j[n] = (\hat{y}_j * h_{\text{lp}})[n] \downarrow f_s'$$

Append signal pair to dataset

$$\mathcal{D}' \leftarrow \mathcal{D}' \cup \left(\tilde{y}_i[n], \tilde{y}_j[n]\right)$$

Append time lag to time-lag set

$$\{\tau\} \leftarrow \{\tau\} \cup \{\tau_r\}$$

end for

return $\mathcal{D}'$, $\{\tau\}$

V. Experimental Setup

A. Training Details

All experiments were conducted on a workstation equipped with an Intel i5-12600KF CPU, 32-GB RAM, and an NVIDIA GeForce RTX 4060 Ti GPU. Model training was performed in PyTorch 2.1.2 using the Adam optimizer (learning rate of 1×10^{-4} and batch size of 32) for 500 epochs. The Smooth L1 (Huber) loss L_κ was used as the training objective in (18), combining the stability of L2 loss with the robustness of L1 loss to outliers. Early stopping was applied, terminating training when the validation loss failed to improve for 300 consecutive epochs. All reported results are averaged over three independent runs to ensure stability

$$L_\kappa(\theta) = \frac{1}{N_t} \sum_{i=1}^{N_t} \begin{cases} \frac{1}{2}(\hat{\tau}_i - \tau_i)^2, & |\hat{\tau}_i - \tau_i| < \kappa \\ \kappa |\hat{\tau}_i - \tau_i| - \frac{1}{2}\kappa^2, & |\hat{\tau}_i - \tau_i| \ge \kappa \end{cases} \tag{18}$$

where κ is the transition parameter, N_t is the number of training samples, $\hat{\tau}_i$ is the predicted lag, and τ_i is the ground truth.

B. Evaluation Metrics

Model performance was quantitatively evaluated using mean absolute error (MAE), root-mean-squared error (RMSE), and the coefficient of determination (R^2) [40]. MAE measures the average magnitude of prediction error by equally weighting all residuals, while RMSE penalizes larger errors more strongly

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due to the squaring operation, making it more sensitive to outliers. The R^2 assesses the proportion of variance in the ground truth explained by the model, with values closer to 1 indicating stronger predictive accuracy. The formulations of MAE, RMSE, and R^2 are given in the following equations:

$$\text{MAE} = \frac{1}{N_t}\sum_{i=1}^{N_t}|\hat{\tau}_i - \tau_i| \tag{19}$$

$$\text{RMSE} = \sqrt{\frac{1}{N_t}\sum_{i=1}^{N_t}(\hat{\tau}_i - \tau_i)^2} \tag{20}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N_t}(\tau_i - \hat{\tau}_i)^2}{\sum_{i=1}^{N_t}(\tau_i - \bar{\tau})^2} \tag{21}$$

where $\bar{\tau} = (1/N_t)\sum_{i=1}^{N_t}\tau_i$ denotes the sample mean of the ground-truth time lags.

C. Baseline Models

To establish strong and representative benchmarks, the proposed TLCA framework was evaluated against both classical FD methods and feature-based machine learning (ML) regressors. This ensures that the evaluation suite reflects well-established analytical techniques as well as competitive data-driven baselines. The FD method estimates the time lag by exploiting the slope of the CSD phase spectrum between two signals [22]. Both signals are first transformed into the frequency domain using the FFT, after which the CSD is computed and its phase spectrum unwrapped. A linear regression is performed on the unwrapped phase with respect to frequency, with the slope providing an estimate of the time lag. To ensure a fair comparison, the FD baseline was tuned over candidate usable frequency bands, FFT window lengths, and overlaps. The configuration achieving the lowest validation MAE was adopted as the optimal FD estimator.

Six widely used ML regressors were included as additional baselines: support vector regression (SVR), random forest (RF), gradient boosting regressor (GBR), extreme gradient boosting (XGBoost), K-nearest neighbors (KNNs), and multilayer perceptron (MLP). SVR maps input features into a high-dimensional space via kernel functions to construct regression functions. RF, GBR, and XGBoost leverage ensemble decision trees to capture nonlinear dependencies. MLP enables flexible nonlinear modeling of complex temporal-amplitude relationships. KNN serves as a simple nonparametric baseline by exploiting local similarities in the feature space. Because these regressors cannot directly handle raw time-series input, hand-crafted features were extracted in both the time and frequency domains. Time-domain features included cross correlation (CC) lag, which estimates temporal misalignment from the correlation peak, as well as statistical descriptors (mean and variance differences), capturing amplitude offsets and discrepancies in signal energy. FD features were derived from the slope of the unwrapped CSD phase spectrum, reflecting the linear relationship between time lag and phase shift across frequencies. All ML baselines were tuned using grid search with fivefold cross-validation on the training set. The search spaces were determined based on prior literature and experiments, and detailed hyperparameter search spaces are provided in Appendix.

TABLE I
ABLATION STUDY OF DIFFERENT MODEL COMPONENTS

Model	MAE	RMSE	R^2 (%)
TimelagCNN only (2,16,32)	32.4	39.7	87.92
ReducedCNN (2,16) + Attention	27.4	21.6	88.38
TimelagCNN + Cross-correlation estimator	7.8	6.6	98.58
TimelagCNN + LSTM	4.7	4.0	98.85
TimelagCNN + Attention	**2.0**	**1.4**	**99.98**

TABLE II
PERFORMANCE OF THE MODEL UNDER DIFFERENT SITUATIONS

Metric	Noise Level (σ)					Modal Coupling
	0	0.02	0.10	0.20	0.50	
MAE	2.0	2.4	3.0	3.9	5.5	6.9
RMSE	1.4	3.4	3.8	3.8	4.3	6.9
R^2 (%)	99.98	99.89	99.86	99.86	99.75	97.64

VI. RESULTS AND DISCUSSION

A. Ablation Study

An ablation study was performed to evaluate the contribution of key components within the TLCA framework (Table I). The baseline CNN yields the largest errors, indicating that standard convolutions alone cannot capture delay-dependent temporal structure. Adding attention to a reduced CNN provides limited improvement, as attention resolves long-range dependencies but cannot compensate for missing lag-aware features. Replacing the attention with a traditional CC estimator or long-short term memory (LSTM) results in competitive performance but still inferior to the full TLCA model. This indicates that fixed analytical methods and recurrent networks lack sufficient resolution for precise delay estimation. The full TLCA model achieves the best performance, confirming that the combination of lag-aware convolutions and attention-based global context modeling is crucial for accurate and robust time-lag prediction. All time-lag error metrics are reported in milliseconds (ms) unless otherwise stated.

B. Robustness Test

To evaluate the robustness, a series of stress tests was conducted under challenging sensing conditions, including increased noise and modal coupling. Gaussian noise with progressively increasing variance was added to the input signals to emulate deteriorating measurement quality, and the model was re-evaluated on each noisy dataset. Modal coupling was introduced by perturbing the modal damping matrix with small off-diagonal terms, generating controlled interactions between modes and representing nonclassical damping commonly observed in real structures. As summarized in Table II, the model exhibits only marginal performance degradation across all scenarios and maintains high predictive accuracy even under severe perturbations. These findings demonstrate the strong resilience of the proposed TLCA framework to input perturbations and confirm its suitability for real-world SHM environments where sensor noise is unavoidable.

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TABLE III
SENSITIVITY ANALYSIS RESULTS OF SIMULATED DATASET

Segment Length	Metric	Time Lag (ms)													
		37.5	50.0	62.5	75.0	87.5	100.0	112.5	125.0	175.0	200.0	250.0	375.0	500.0	625.0
1000	MAE	0.34	0.68	0.59	0.60	0.60	0.66	0.77	0.65	0.99	1.38	2.01	3.15	4.35	9.80
	RMSE	0.48	0.92	0.75	0.91	0.87	0.97	1.07	1.09	1.28	1.82	2.89	4.60	8.34	26.47
	R^2 (%)	99.93	99.85	99.94	99.93	99.96	99.96	99.96	99.97	99.98	99.96	99.94	99.93	99.87	99.14
2000	MAE	0.27	0.38	0.34	0.81	0.35	0.64	0.59	0.47	0.76	1.27	1.39	2.16	2.23	3.74
	RMSE	0.35	0.46	0.44	0.96	0.46	0.78	0.71	0.60	1.01	1.54	1.81	2.77	2.99	6.92
	R^2 (%)	99.96	99.96	99.98	99.93	99.99	99.97	99.98	99.99	99.98	99.97	99.98	99.98	99.98	99.94
3000	MAE	0.21	0.26	0.35	0.40	0.36	0.44	0.92	0.63	0.82	0.76	1.23	2.10	3.69	3.55
	RMSE	0.27	0.33	0.43	0.51	0.46	0.55	1.06	0.77	1.04	0.95	1.57	2.71	4.82	4.71
	R^2 (%)	99.98	99.98	99.98	99.98	99.99	99.99	99.96	99.98	99.98	99.99	99.98	99.98	99.96	99.97

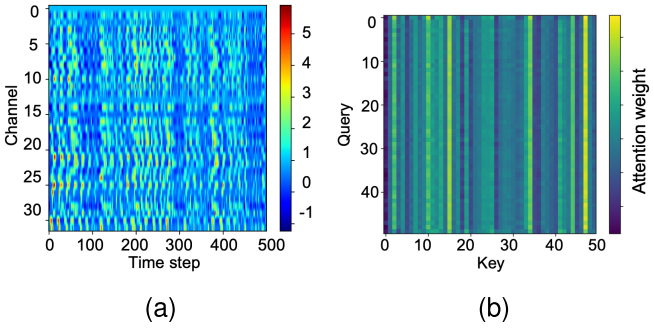


Fig. 8. (a) Feature map after convolutional layers. (b) Attention map.

C. Model Complexity and Efficiency

The proposed TLCA model contains 108k parameters and occupies 0.41 MB of memory. This lightweight architecture enables real-time inference, achieving end-to-end latencies of 0.04 ms on a GPU and 0.14 ms on a CPU, which satisfies the timing requirements of vibration-based SHM applications. A pair of 2000-point input segments require 3.87 million FLOPs and corresponds to a message size of 31.2 kB, making the model suitable for bandwidth-constrained SHM IoT systems. The full training process completes in approximately 4 min on the GPU and 12 min on the CPU. These results indicate that TLCA is computationally affordable and well suited for deployment in resource-constrained SHM IoT nodes.

D. Feature Visualization

To further examine the internal representation learned by the proposed model, the convolutional feature maps and the corresponding self-attention weights are visualized. The feature map exhibits well-structured temporal activations across channels [Fig. 8(a)], indicating that the convolutional layers successfully extract discriminative time–frequency patterns from the input sequence. Clear periodic responses and localized energy concentrations are preserved, suggesting that both short-term fluctuations and longer-range temporal characteristics are well encoded. The attention map in Fig. 8(b) displays sharp and localized peaks across key positions, demonstrating that the self-attention mechanism selectively emphasizes informative temporal regions that are most relevant to time-lag estimation.

E. Sensitivity Analysis Results

The sensitivity analysis was conducted to assess the impact of segment length and time-lag range on model performance. The results for the simulated and field datasets are summarized in Tables III and IV, respectively.

1) Simulated Dataset: For the simulated structural responses, TLCA maintained consistently high accuracy across a wide range of time lags (37.5–625 ms). As shown in Table III, increasing the segment length led to a clear reduction in MAE and RMSE, particularly for larger time lags. With the segment of 3000, the MAE and RMSE remained below 1 ms for time lags up to 200 ms and below 4 ms even at the maximum lag of 625 ms. R^2 values approached 1.0 across most configurations, indicating that the model reliably captures the underlying phase–time relationship. Performance degradation becomes noticeable only for very large delays combined with short segments, reflecting the reduced temporal context available for estimation.

2) Field Dataset: The sensitivity trends observed in the field dataset (Table IV) were consistent with those of the simulated data, albeit with higher absolute errors due to environmental noise and structural complexity. For the field data, clearly abnormal or corrupted measurements were identified and removed in advance to avoid biasing the subsequent analysis. For all tested segment lengths and maximum lag ranges up to 400 ms, TLCA preserved strong predictive performance, with MAE typically below 6 ms and RMSE below 9 ms. Increasing the segment length generally improves robustness, particularly under larger maximum lag conditions. Notably, the R^2 values remain above 99.9% across nearly all settings, confirming stable generalization under real-world sensing conditions.

F. Comparison Results With Baseline Models

The proposed TLCA model was evaluated against FD methods and advanced ML regressors on both simulated and field datasets, as summarized in Table V. On the simulated dataset, the time-lag range was set to 500 ms, representing a practical range. All ML-based baselines outperform the FD method, demonstrating the advantage of data-driven regression over analytical phase-slope estimation under controlled conditions. Among the ML models, ensemble-based methods (RF, GBR,

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TABLE IV
SENSITIVITY ANALYSIS RESULTS OF FIELD DATASET (MAE/RMSE IN MS AND R^2 IN %)

Segment Length	Metric	Max Lag (ms)									
		40	80	120	160	200	240	280	320	360	400
1000	MAE	0.68	1.81	3.99	3.65	6.42	4.15	5.56	5.44	5.15	8.91
	RMSE	1.35	2.80	5.30	4.92	8.77	6.02	8.44	8.73	7.67	11.08
	R^2 (%)	99.99	99.99	99.98	99.99	99.98	100.00	99.99	99.99	99.99	99.99
1600	MAE	0.95	1.29	7.92	3.03	3.12	2.35	5.14	7.28	4.10	6.49
	RMSE	1.33	2.26	8.65	4.99	6.62	3.79	6.71	9.60	6.49	7.16
	R^2 (%)	99.94	99.99	99.90	99.95	99.99	99.997	99.99	99.99	99.99	99.99
2000	MAE	0.91	1.70	3.14	3.03	3.61	2.51	3.59	4.81	4.08	5.01
	RMSE	1.33	2.15	5.64	4.02	6.82	3.01	5.03	5.54	5.09	7.01
	R^2 (%)	99.99	99.99	99.98	99.99	99.99	99.99	99.99	100.00	99.99	99.97

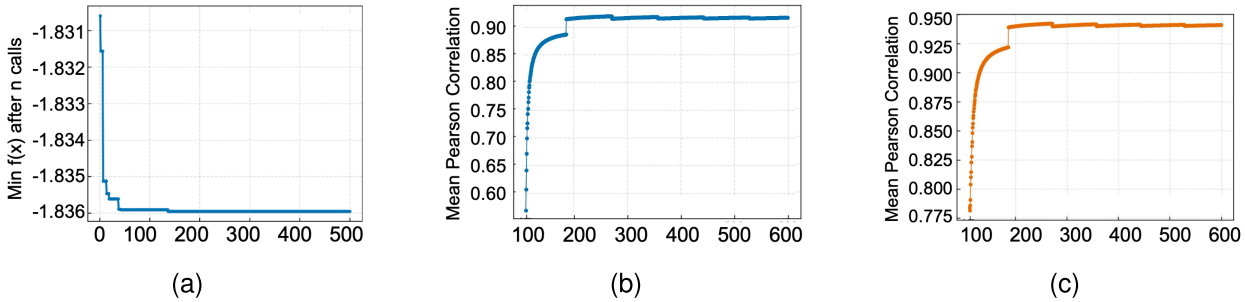


Fig. 9. (a) Convergence curve of Bayesian optimization. (b) Mean Pearson correlation versus upsampling frequency. (c) Max cross correlation upsampling frequency.

TABLE V
COMPARISON RESULTS AMONG DIFFERENT MODELS

Model	Simulated Dataset			Field Dataset		
	MAE	RMSE	R^2(%)	MAE	RMSE	R^2(%)
FD	28.6	37.1	98.74	122.2	133.7	0.37
SVR	11.1	12.7	99.85	119.4	132.9	0.13
RF	10.4	12.9	99.85	45.6	76.6	66.81
GBR	9.9	11.7	99.88	50.5	76.4	67.03
XGBoost	9.9	11.7	99.88	52.6	77.6	65.92
KNN	10.0	13.4	99.84	119.8	149.1	25.71
MLP	14.0	16.6	99.75	45.6	76.6	66.81
TLCA	**2.0**	**1.4**	**99.98**	**5.0**	**7.3**	**99.97**

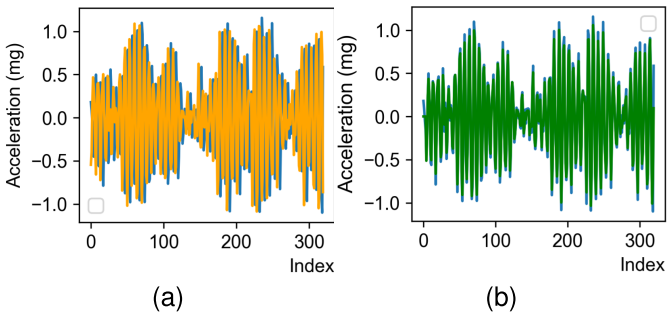


Fig. 10. Synchronization results of simulated dataset. (a) Original (blue) versus lagged signal (orange). (b) Original (blue) versus aligned signal (green).

and XGBoost) achieve comparable performance, with MAE values on the order of 10 ms. TLCA significantly outperforms all baselines, achieving an MAE of 2 ms and an RMSE of 1.4 ms. The near-unity R^2 further confirms the high accuracy of TLCA in estimating continuous-valued time lags.

The performance gap was further amplified on the field dataset with a time-lag range of 400 ms, which is characterized by stronger noise, environmental variability, and modeling uncertainties. TLCA consistently achieved millisecond-level errors with an MAE of 5 ms, an RMSE of 7 ms, and an R^2 of 99.97%, maintaining high accuracy and robustness. The FD method and kernel-based regressors (SVR and KNN) fail to generalize, yielding large errors and extremely low R^2 values, indicating poor explanatory power. Although ensemble ML models and the MLP show moderate improvements, their performance remains limited, with MAE values exceeding 45 ms and R^2 below 70%.

G. Synchronization Results

Once the time lags were estimated, signals were subsequently synchronized using the resampling method with BO, with the upsampling frequency searched over 100–600 Hz for 500 iterations, and the convergence curve shown in Fig. 9(a). The best performance was achieved at 340 Hz, where both correlation metrics reached their peak values, as shown in Fig. 9(b) and (c). Beyond this point, further increases in frequency yielded negligible improvement, indicating saturation of alignment performance. This procedure enables precise subsample alignment with spectral fidelity while balancing computational efficiency with synchronization accuracy.

The synchronization performance on the simulated and field datasets is illustrated in Figs. 10 and 11, respectively. For the simulated dataset, Fig. 10(a) compares the ground-truth response from Sensor 2 with its artificially delayed

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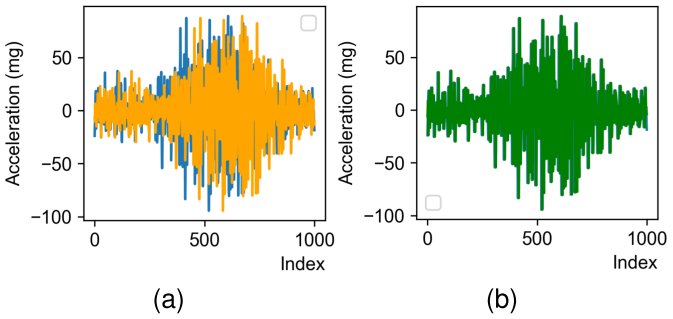


Fig. 11. Synchronization results of field dataset. (a) Original (blue) versus lagged signal (orange). (b) Original (blue) versus aligned signal (green).

counterpart, clearly showing temporal misalignment. After applying the BO-assisted resampling method, the synchronized signal exhibits strong overlap with the reference response, as shown in Fig. 10(b). The alignment effectively corrects the imposed noninteger time lag while preserving the waveform structure and phase consistency.

For the field dataset collected from the Jindo Bridge, Fig. 11(a) shows the asynchronous signal pair, where a clear negative time offset is observed due to unsynchronized sensing across sensor nodes. Using the same BO-assisted resampling framework with the search space of BO set from 21 to 500 Hz over 500 iterations, the signals were realigned with the optimal upsampling frequency determined by BO at 125 Hz. As shown in Fig. 11(b), the synchronized signal closely matches the reference measurement, demonstrating that the proposed approach remains effective under realistic field conditions characterized by noise, environmental variability, and modeling uncertainty.

A slight amplitude bias arised from nonideal passband and transition-band characteristics FIR filter used in resampling [34]. This occurs because resampling lowers the Nyquist frequency, requiring attenuation of high-frequency components to prevent aliasing. Since most downstream SHM tasks rely on these low-frequency characteristics, the impact of resampling-induced filtering is negligible and may even improve robustness by suppressing high-frequency noise. Nevertheless, this represents an inherent limitation of resampling-based synchronization. For applications involving significant high-frequency dynamics, filter design parameters should be carefully tuned.

H. Impact on Downstream SHM Tasks

Accurate time synchronization is critical for vibration-based SHM tasks of modal identification and damage assessment. To quantify the impact of synchronization quality on downstream analysis, modal parameters and mode shapes were identified from both asynchronous and aligned datasets and compared with analytical results. The theoretical modal parameters and mode shapes were obtained from the state space model by solving the generalized eigenvalue problem $\mathbf{K\Phi} = \mathbf{M\Phi\Lambda}$. Modal parameters and mode shapes from asynchronous and aligned datasets were identified by the eigensystem realization algorithm (ERA). The results comparison are summarized in Table VI and Fig. 12. It shows that natural frequencies are unaffected by temporal misalignment and damping ratios

TABLE VI
MODAL PARAMETERS OF DOWNSTREAM SHM TASKS

Mode	Frequency (Hz)			Damping ratio %			MAC	
	Theory	Sync	Async	Theory	Sync	Async	Sync	Async
1	11.192	11.192	11.072	2.18	2.06	1.85	0.9996	0.4758
2	30.167	30.167	30.012	3.31	3.19	3.07	0.9997	0.0080
3	47.315	47.315	47.293	1.94	1.85	1.75	0.9996	0.8697
4	59.804	59.804	59.803	4.06	3.95	3.41	0.9997	0.7157

TABLE VII
COMPARISON AMONG DIFFERENT SYNC METHODS

Method	MSC	MCC	Time (s)
Index shift (integer)	0.93	0.87	0.01
Linear interpolation	0.95	0.90	1.12
Resampling enumeration	0.99	0.97	12.40
BO-assisted resampling	**0.98**	**0.97**	**1.14**

exhibit minor variations. The mode shapes identified from asynchronous data exhibit significant distortion and fail to align with the theoretical patterns, as reflected by very low modal assurance criterion (MAC) values. Such distortions can lead to incorrect modal interpretation and unreliable damage indicators. After signal alignment, the mode shapes closely match the theoretical model, achieving MAC values near 1.0 across all modes.

I. Comparison With Other Synchronization Methods

To quantitatively evaluate the effectiveness of the proposed BO-assisted resampling method, it was compared against the traditional index-shift method, linear interpolation, and resampling enumeration without BO. Performance was assessed using three complementary metrics: magnitude spectrum correlation (MSC), maximum CC (MCC), and computational time. MSC quantifies spectral fidelity by measuring the similarity of amplitude spectra between the aligned and reference signals. MCC evaluates time-domain alignment quality by capturing waveform similarity after synchronization. The results are summarized in Table VII The index-shift method approximates noninteger delays as integer shifts and yields the lowest MCC, indicating poor correction of subsample misalignment. Linear interpolation provides modest improvement but remains limited by fixed interpolation kernels. Exhaustive resampling achieves high alignment accuracy, as reflected by high MSC and MCC values but incurs a prohibitive computational cost. In contrast, the proposed BO-assisted resampling method achieves comparable spectral and temporal alignment accuracy to exhaustive resampling while reducing computation time by nearly an order of magnitude. These results demonstrate that BO enables efficient subsample synchronization with high spectral fidelity, without the computational burden of brute-force resampling.

J. Discussion

In wireless SHM systems, network-level time synchronization and post hoc signal alignment play complementary roles in achieving reliable synchronization and can be combined in practice. Network-based synchronization protocols (e.g., FTSP and TPSN) provide coarse alignment by ensuring consistent timestamps across nodes but cannot ensure

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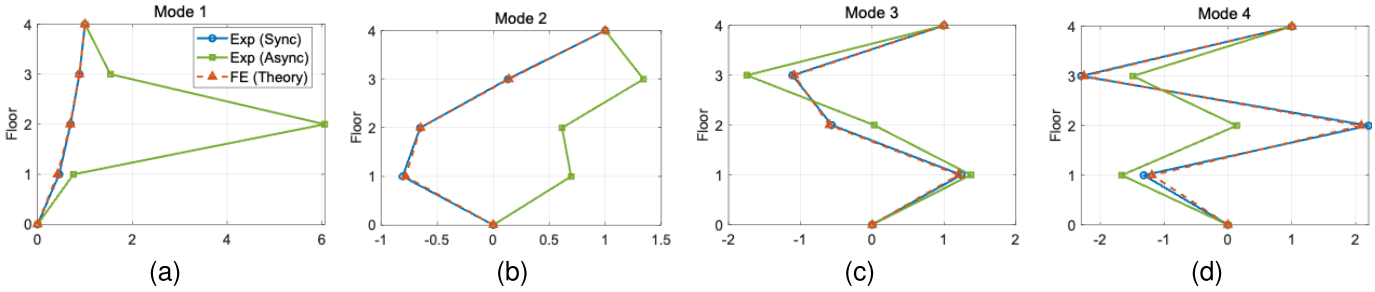


Fig. 12. Mode shape comparison across sync, async, and theory conditions. (a) Mode1. (b) Mode2. (c) Mode3. (d) Mode4.

synchronized sensing because of nondeterministic delays in sensing and data acquisition. Signal-level synchronization is to provide fine-grained correction and eliminate these remaining misalignments during postprocessing. The proposed work is positioned at the post hoc signal-level synchronization to estimate time lags directly from vibration data and correct the residual offsets that remain after network-level synchronization. It performs learned fine alignment, enabling high-precision synchronization for vibration-based SHM applications.

VII. Conclusion

This study introduced a novel end-to-end DL-based framework for data synchronization in wireless SHM systems. The proposed TLCA network directly estimates time lags from raw vibration data, eliminating reliance on handcrafted features and restrictive signal assumptions. Additionally, a resampling strategy enhanced with BO was developed to address noninteger time lags, ensuring synchronized signals that preserve critical modal information. Validation on both simulated responses and field data confirmed that TLCA achieves superior accuracy, robustness to noise, and scalability. Compared with the conventional FD method and ML models, TLCA consistently delivered better performance across all evaluation metrics. The motivation for adopting TLCA lies in the strong representational capacity of DL models, which are not restricted by simplified analytical assumptions and user experience. The superior performance of TLCA arises from the joint use of convolutional and attention mechanisms, which enable the model to learn discriminative representations directly from raw vibration time-series data and to selectively emphasize temporal features most relevant to time-lag estimation. In contrast, traditional FD and feature-based methods rely on a limited set of handcrafted features and relatively simple learning models, which fundamentally restrict their ability to resolve continuous-valued, subsample time lags and thus limit synchronization accuracy. The integration of resampling with BO enabled precise correction of subsample delays without the need for timestamps, outperforming traditional resampling methods. The framework improves the reliability of downstream tasks of modal identification and damage detection, thereby supporting safer and more resilient infrastructure monitoring. A limitation of the proposed resampling strategy is the minor amplitude bias introduced by finite-length FIR filters, and future work will further optimize the filter design to minimize such effects and explore alternative synchronization strategies that balance accuracy, computational efficiency, and signal integrity. Furthermore, the framework will be extended to explicitly account for nonuniform sampling effects, including sampling jitter, clock drift, and heterogeneous sampling rates across sensor nodes, as well as mild nonlinear dynamics, to enable robust synchronization under more realistic operating conditions. In addition, we will investigate the real-world deployment of the proposed approach on actual edge hardware, including benchmarking on physical devices and evaluating performance and reconstruction quality under real operating conditions.

Appendix
Hyperparameter Search Space for Baseline ML Models

The hyperparameter search space for baseline ML models is summarized as follows.

1) *SVR:*
   a) Kernel type $\in$ {RBF, poly}.
   b) Regularization parameter $C_r \in \{0.1, 1, 10, 100\}$.
   c) $\gamma \in \{\text{scale}, \text{auto}, 0.01, 0.1, 0.25, 0.5, 1\}$.
   d) $\epsilon \in \{0.001, 0.01, 0.1, 0.2, 0.5\}$.
   e) Here, γ denotes the kernel coefficient and ϵ defines the width of the ϵ-insensitive region.
2) *RF:*
   a) $n_{\text{estimators}} \in \{50, 100, 200, 400\}$.
   b) max_depth $\in \{10, 20, 30, \text{None}\}$.
   c) min_samples_split $\in \{2, 5, 10\}$.
   d) min_samples_leaf $\in \{1, 2, 4\}$.
   e) max_features $\in$ {auto, sqrt, log2}.
3) *GBR:*
   a) $n_{\text{estimators}} \in \{50, 100, 200, 400\}$.
   b) learning_rate $\in \{0.01, 0.05, 0.1\}$.
   c) max_depth $\in \{3, 5, 7\}$.
   d) min_samples_split $\in \{2, 5, 10\}$.
   e) min_samples_leaf $\in \{1, 2, 4\}$.
   f) max_features $\in$ {auto, sqrt, log2}.
   g) subsample $\in \{0.6, 0.8, 1.0\}$.
4) *XGBoost:*
   a) $n_{\text{estimators}} \in \{50, 100, 200, 400\}$.
   b) learning_rate $\in \{0.01, 0.05, 0.1\}$.
   c) max_depth $\in \{3, 5, 7\}$.
   d) min_child_weight $\in \{1, 3, 5\}$.
   e) Kernel coefficient$\gamma \in \{0, 0.1, 0.3\}$.
   f) subsample $\in \{0.6, 0.8, 1.0\}$.
   g) colsample_bytree $\in \{0.6, 0.8, 1.0\}$.

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5) *KNN:*
   a) $n_{\text{neighbors}} \in \{3, 5, 7, 9, 11, 15\}$.
   b) weights $\in$ {uniform, distance}.
   c) Distance metric parameter $p_d \in \{1, 2, 3, 4, 5\}$.
6) *MLP:*
   a) Hidden layers $\in$ {(32), (32, 16), (64), (64, 32),. (128, 64), (128, 64, 32), (256, 128)}.
   b) Activation $\in$ {ReLU, tanh}.
   c) Learning rate $\in \{10^{-3}, 10^{-4}\}$.

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